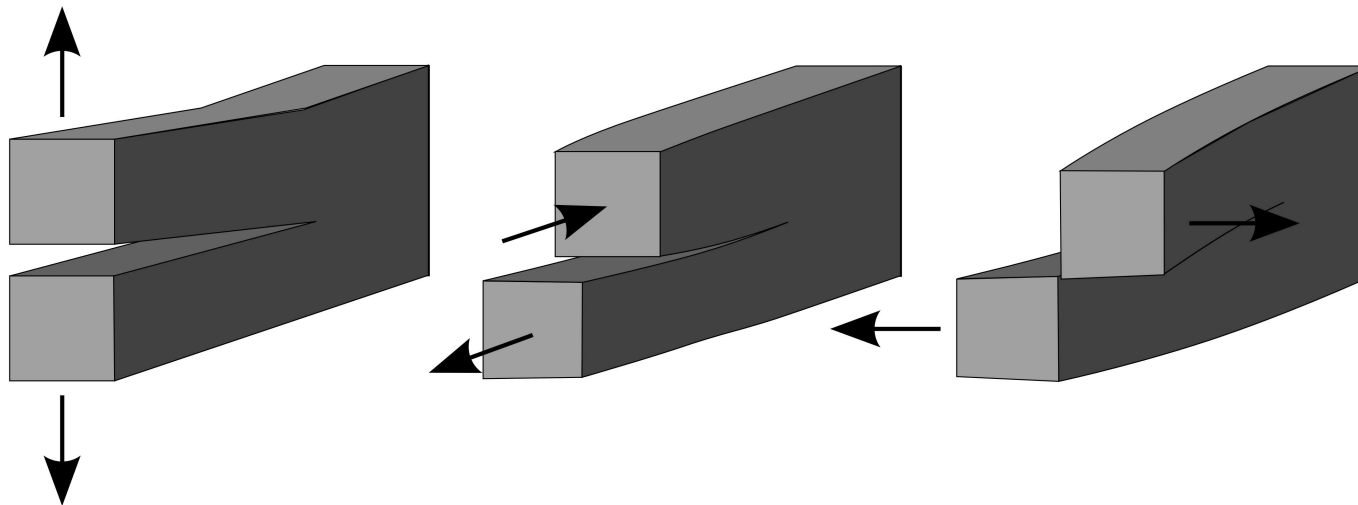


George Rankine Irwin's Approach (1957)

Although the load on a crack can be arbitrary, it was found any state could be reduced to a combination of three independent stress intensity factors:

- Mode I - Opening mode (a tensile stress normal to the plane of the crack),
- Mode II - Sliding mode (a shear stress acting parallel to the plane of the crack and perpendicular to the crack front), and
- Mode III - Tearing mode (a shear stress acting parallel to the plane of the crack and parallel to the crack front).



Mode I:
Opening

Mode II:
In-plane shear

Mode III:
Out-of-plane shear